Yu Zhang

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Research Interests

Large Language Models (LLMs), Natural Language Processing (NLP) with Reinforcement Learning (RL), Multimodal NLP, Interdisciplinary Clinical NLP

Education

Cornell University | Ithaca, US *Aug 2025 – Dec 2026 (Expected)*

M.Eng. Systems Engineering | GPA: 3.96/4.0

Core Modules: AI Reasoning & Decision-Making, Clinical NLP & LLMs, Natural Language Processing, Advanced Machine Learning (LLMs)

University of Southampton | Southampton, UK *Sep 2021 – Sep 2022*

M.Sc. Artificial Intelligence | GPA: 3.5/4.0

Core Modules: Intelligent Agents, Artificial Intelligence, Machine Learning, Advanced Machine Learning, Computer Vision, Reinforcement and Online Learning, Deep Learning

Xi'an Jiaotong-Liverpool University &University of Liverpool | Suzhou, China & Liverpool, England *Sep 2016 – Jul 2020*

B.Eng. Computer Science and Electronic Engineering

Core Modules: Database Development, Programming in Java, C++ Programming, Software Engineering

Research Experience

Generative Multimodal Intelligence for Sustainability Audit | Cornell University, US

Position: Team Leader *Sep 2025 – Present*

- Built end-to-end PDF processing pipeline using Docling and spaCy to extract quantitative and qualitative indicators from corporate sustainability reports
- Constructed a unified multimodal dataset containing text and numerical data to train Generative Adversarial Networks (GANs) and Transformer models for sustainability benchmarking
- Designed a "Trust RAG" workflow to evaluate corporate sustainability practices across multiple dimensions and metrics
- Refined prompt engineering for ChatGPT, Doubao, Gemini, Claude, and used evidence-based scoring to rank and compare firms
- Fine-tuned AI agents on sustainability reports and compared the results against those from prompt engineering
- Planned web front-end integration to support report uploads and real-time summaries

Machine Learning-based Text Generation | Heilongjiang Artificial Intelligence Institute, China

Position: Team Leader *Nov 2022 – Feb 2023*

- Led a research project analyzing 60+ social media topics (500 posts, 3k comments each) to advance neural text generation
- Designed a modular pipeline combining LDA, SVR and BERT, contributing to a peer-reviewed publication on contextual content synthesis
- Developed an emotion- and topic-aware comment generator with PLSA and ConPRG, improving diversity and contextual relevance

Control of Repetitive Tasks using Reinforcement Learning | University of Southampton, England

Position: Researcher Mar 2022 – Sep 2022

- Modeled second-order dynamic systems using iterative learning control (ILC) and norm-optimal ILC in MATLAB
- Integrated Q-table into classical ILC structures and conducted controlled experiments
- Built a neural-network-free Actor-Critic ILC framework and verified its convergence on a simulated printer control system
- Tuned core hyperparameters to benchmark algorithm performance under diverse dynamic conditions

BERTScore Reproduction for Text Generation Evaluation | University of Southampton, England

Position: Team Leader *Mar 2022 – May 2022*

- Rebuilt the BERTScore evaluation pipeline using multilingual WMT18 corpus subsets across five languages
- Applied pre-trained models (BERT, RoBERTa, XLNet, XLM) for cross-lingual machine translation and semantic alignment
- Verified result reproducibility and measured variance against the original paper

Deep Reinforcement Learning for StarCraft II Gameplay | University of Liverpool, England

Position: Researcher *Sep 2019 – May 2020*

- Simulated StarCraft II environments with 6 Protoss structures, 7 weapons and 16 random actions via the official API
- Integrated A3C and PPO to build a distributed PPO network for asynchronous policy updates across 4 parallel environments
- Trained agents over 400 iterations, achieving 73%/51%/39% win rates against simple/medium/advanced AI and validating policy generalization

Professional Experience

BinXian Zeqiu Xiaoquan Fresh Supermarket | China

Position: Co-Founder & Engineer *Sep 2023 – Jul 2025*

- Oversaw 5 supermarket locations with 80K RMB ($15K) daily revenue, 600+ daily customers, and 20K+ SKUs
- Streamlined procurement and supplier onboarding, deployed data-driven membership campaigns, developed dynamic pricing models, and automated inventory tracking to cut discrepancies by 20%
- Managed ERP and database systems, re-built the 20K+ SKU supply chain databases, and rolled out digital membership systems
- Built automated invoicing tools that handled 300K+ RMB monthly, reducing workload by 70%

DataMIMO | Remote, US

Position: Data Scientist (Volunteer) *Jun 2023 – Aug 2023*

- Automated data extraction from Morningstar and Macrotrends using Python
- Designed multithreaded C++ pipelines that lowered data retrieval latency by 24%
- Implemented quantitative optimization and Random Forest models, lifting the Sharpe ratio by 0.1

Suzhou Speedlight| China

Position: Data Analytics Intern *Mar 2021 – Jul 2021*

- Analyzed multi-domain data streams using packet-capture techniques
- Collected large-scale datasets and implemented ad-targeting segmentation
- Developed an adaptive data capture system with compression algorithms

Publications

- Y. Zhang, “Incorporating emotional trend into multi-emotion analysis models for long-text sentiment analysis,” in Proc. Int. Conf. Machine Learning and Applications (ICMLA), 2023, doi: 10.54254/2755-2721/44/20230241.

- Z. Yang, H. Xie, Y. Xu, Q. Xu, Y. Zhang, S. Zhao, H. Yuan, and Y. Fang, “Evaluation of smart response systems for city emergencies and novel UV-oriented solution for integration, resilience, inclusiveness and sustainability,” in Proc. 5th Int. Conf. Universal Village (UV), 2020, doi: 10.1109/UV50937.2020.9426215.
- Y. Zhang, “Application and development of computer artificial intelligence technology,” Journal of Computer Science Research, 2019, ISSN: 2630-5151

Awards & Extracurricular

- MIT Universal Village Summer Research (Team Lead, Emergency System)
- University Entrepreneurship Competition (3rd Prize, “Word Arena” project)
- Bali Marine Conservation Volunteer

Technical Skills

- **Languages**: Mandarin (Native), English (Fluent)
- **Programming**: Python, C++, Java, R, MATLAB
- **Tools**: PyTorch, TensorFlow, Scikit-learn, Keras